HINTS & SOLUTOIN

Q.1 (A)

Q.2 (C)

Q.3 (A)

$$dN = 4\pi N\left(\frac{M}{2\pi RT}\right)^{3/2} 4^2 e^{-\frac{Mn^2}{2RT}du}$$

$$\frac{dN}{du}\,\alpha\, N\left(\frac{M}{T}\right)^{3/2} 4^2 e^{-\frac{Mu^2}{2RT}}$$

$\frac{M}{T}$ for both is same

N for O_2 > N for SO_2

So (A) Ans.

Q.4 (B)

$$V_c = \frac{30}{0.40} \Rightarrow 75\ cm^3mol^{-1}$$

$$\therefore\ b = \frac{V_c}{3} \Rightarrow 25\ cm^3mol^{-1} \Rightarrow 0.025\ Lmol^{-1}$$

$$\because\ T_c = \frac{8a}{27Rb}$$

$$\frac{2\times10^5}{821} = \frac{8\times a}{27\times0.0821\times0.025} \Rightarrow a = 1.6875$$

Q.5 (B)

$$T_B = \frac{a}{bK} = \frac{3.284}{0.1\times0.0821} = 400K$$

Z = 1

$$Z = 1 + \left(b - \frac{a}{RT}\right)\frac{1}{V_m} + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + \frac{b^4}{V_m^4} + \ldots\ldots\ldots$$

$$Z = 1 + \frac{1/16}{1-1/4} = 1 + \frac{1/16}{3/4} = \frac{13}{12}$$

Q.6 (C)

Q.7 (C)

At very high pressure

$$Z = 1 + \frac{Pb}{RT}$$

$$\frac{dZ}{dp} = \frac{b}{RT} = \frac{1}{10}\ atm^{-1}$$

$$b = \frac{22.4}{273} \times \frac{273}{10} = 2.24\ litre/mole$$

$$b = 4 \times V \times N_A$$

$$V = \frac{2.24 \times 10^3}{4 \times 6 \times 10^{23}} \Rightarrow 9.3 \times 10^{-22}\ cc$$

Q.8 (A)

$$\frac{r_{SO_2}}{r_{He}} = \frac{p_{SO_2}}{p_{He}}\sqrt{\frac{M_{He} \times T_{He}}{M_{SO_2} \times T_{SO_2}}} \Rightarrow \frac{n}{r_{He}} = \frac{1}{0.25}\sqrt{\frac{4}{64} \times \frac{2 \times 273}{273}}$$

$$\frac{n}{r_{He}} = \sqrt{2} \qquad r_{He} = \frac{n}{\sqrt{2}}$$

Q.9 (B)

Real gas deviates from ideal behaviour at high pressure as volume occupied by molecules and interactive forces among molecules are not negligible.

Q.10 (B)

At normal boiling point K.E. (liq. phase) = K.E. (Vapour phase)]

Q.11 (A)

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$$P = \frac{RT}{V - b} - \frac{a}{V^2}$$

$$Z = \frac{PV}{RT} = \frac{V}{V - b} - \frac{a}{VRT}$$

$$= \frac{V}{V - b} - \frac{b}{V} \qquad \left(T = \frac{a}{Rb}\right)$$

$$= 1 + \frac{b^2}{V(V - b)}$$

Q.12 (C)

Q.13 (D)

(A) At T = 500 K, P = 40 atm, the state will be gas

(B) At T = 300 K, P = 50 atm, the state will be liquid

(C) At T < 300 K, P > 20 atm, the state will be liquid

(D) At 300 K < T < 500 K, P > 50 atm, the state will be liquid.

Q.14 (B)

At same temperature, $U_{rms}\ \alpha\ \dfrac{1}{\sqrt{m}}$

Q.15 (D)

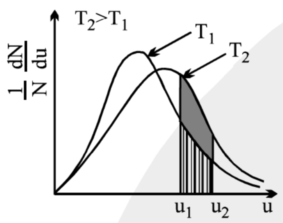
If temperature of gas A in flask-1 & flask-2 are T_1 & T_2 respectively. Finally temperature of both flask is in between T_1 & T_2

Q.16 (A) (B) (C)

(A) $U_{avg} = \sqrt{\dfrac{8}{\pi}\dfrac{RT}{M}}$; So $M \uparrow = U_{avg} \downarrow$

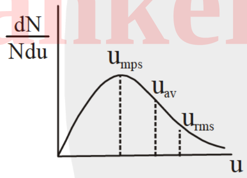
So Momentum decreases

(B)



(C) Average relative speed of approach $= \sqrt{2}u_{avg}$.

(D)



Q.17 (B) (D)

Let T°K ⇒ (t°C + A)K

$PV_m = R(t°C + A)$

$20 = R\,(30 + A)$

$40 = R\,(300 + A)$

$$40 = \frac{20(300 + A)}{(30 + A)}$$

$60 + 2A = 300 + A$

$A = 240$

$R = 0.0741$ L-atm/K-mol

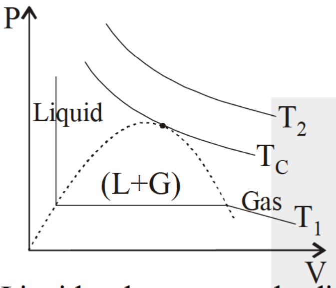
Q.18 (A) (B) (C) (D)

(A) $Z = \dfrac{P_C V_C}{RT_C} = \dfrac{3}{8}$

$Z < 1$

attractive forces are dominating

(B)



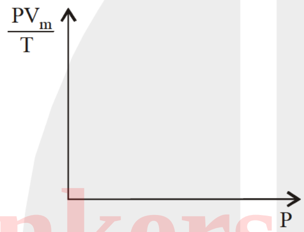
Liquid and gas can not be distinguished above critical temperature.

(C) At very high pressure , $Z > 1$

$\Rightarrow \dfrac{V_m(\text{real})}{V_m(\text{ideal})} > 1$

$\Rightarrow V_m(\text{real}) > V_m(\text{ideal})$

(D) For y-Intercept, X = 0



When $P \rightarrow O$

Gas shows ideal behaviour

$PV_m = RT$

$\Rightarrow \quad \dfrac{PV_m}{T} = R$

$\Rightarrow \quad y = R$

Q.19 (A) (B) (C) (D)

Q.20 (B) (C)

(i) n =2, $T_B = 300$ K ; $\quad P_B = \dfrac{nRT}{V} = \dfrac{2 \times R \times 300}{8} = \dfrac{600}{8} R$

$V_B = 8$ L $\quad = \dfrac{150}{2} R$

(ii) $\quad P_C = \dfrac{nRT}{V_C} = \dfrac{2 \times R \times 300}{10} = 60$ R

(iii) $T_A = \dfrac{P_A V_A}{nR} = \dfrac{2 \times 8}{2R} = \dfrac{8}{R}$

(iv) $T_D = \dfrac{P_D V_D}{nR} = \dfrac{2 \times 10}{2 \times R} = \dfrac{10}{R}$

Q.21 (A) S; (B) R, (C) Q; (D) P

Q.22 (A) P,R,S (B) P,Q,S (C) P,Q,S (D) P,R,S

Q.23 0114

(i) $z_{11} = \frac{1}{\sqrt{2}}\pi\sigma^2 u_{av} N^{*2}$, $PV = \left(\frac{N}{N_A}\right)RT \Rightarrow \frac{N}{V} = \frac{P}{KT}$

$$\frac{(z_{11})_A}{(z_{11})_B} = \left(\frac{0.16}{0.32}\right)^2 \times \left(\sqrt{\frac{1600}{200}\times\frac{40}{20}}\right) \times \left(\frac{8}{1}\right)\left(\frac{200}{1600}\right)$$

$$= \frac{1}{4}\times 4\times\left(\frac{8}{2}\times\frac{200}{1600}\right) = 1$$

(ii) $z_1 = \sqrt{2}\ \pi\,\sigma^2 u_{av} N^*$

$$\frac{(z_1)_A}{(z_1)_B} = \left(\frac{0.16}{0.32}\right)^2 \times 4 \times \frac{8}{1}\times\frac{200}{1600} = 1$$

(iii) $$\frac{(u_{rms})_A}{(u_{rms})_B} = \sqrt{\frac{1600}{200}\times\frac{40}{20}} = 4$$

Q.24 8

$$\left(P + \frac{a}{V_m^2}\right)V_m = RT$$

$$PV_m + \frac{a}{V_m} = R\times\frac{8}{0.0821}$$

$$PV_m^2 - 8V_m + a = 0$$

At any pressure gas can have only one molar volume.

$64 - 4 \times ap = 0$

$$P = \frac{64}{4\times 2} = 8 \text{ atm}$$

Q.25 1836

Initial pressure

	Chamber I	**Chamber II**
T	$N_2O_4 = 20$ mm	H_2O V = 20 mm
1.2 T	$N_2O_4 = 24$ mm	V = 30 mm

$\downarrow$

$P_{N_2O_4}$ rem = 12 mm   30 + 6

$P_{NO_2} = 24 / 4 \Rightarrow 6$ 36

$12 + 6 \Rightarrow 18$

Q.26 6

$$D_m(g) \rightleftharpoons mD(g)$$

$\quad\quad\quad\quad\quad\quad 1 \quad\quad\quad\quad m$

After reaction $\;0.4 \quad\quad\quad 0.6\text{ m}$

$$\frac{n_{actual}}{n_{expected}} = \frac{0.4 + 0.6\text{m}}{1} \quad \Rightarrow \quad 0.4 + 0.6\text{ m} = \frac{1.2}{0.3} \quad \Rightarrow \quad 0.4 + 0.6\text{ m} = 4$$

$\Rightarrow \quad 0.6\text{ m} = 3.6 \quad \Rightarrow \quad m = 6$

Q.27 0775

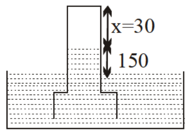
$x \times 10 = (x - 20)\,30 = 20 \times P$

$x = 30$

$20 \times P = 30 \times 10$

$P = 15$

$P_T = 15 + 760 = 775$



Q.28 0025

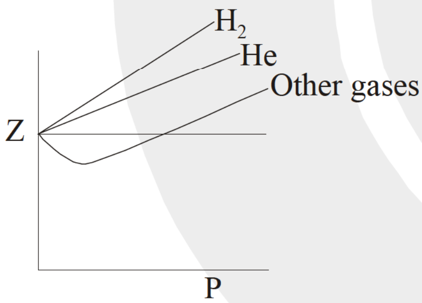
Mass of water vapour in 10L at 27°C $= M\left(\dfrac{P_1V}{RT_1}\right) = 18\times\left(\dfrac{25\times10\times0.80}{760\times0.0821\times300}\right) = 0.192\text{ gm}$

Mass of water vapour in 10L at 7°C $= M\left(\dfrac{P_2V}{RT_2}\right) = 18\times\left(\dfrac{14\times10}{760\times0.0821\times280}\right) = 0.14424\text{ gm}$

Mass % of water which will condonse $= \left(\dfrac{0.192 - 0.14421}{0.2404}\right)\times 100 = 25\%$

Q.29 4 : 7 : 5

Q.30 0005



Normal Temperature

(1) $Z > 1$ (from graph)

(2) At $T < T_C$, attraction will be present and for He and H_2 also $Z < 1$

(3) $Z > 1$ (from graph)

(4) $Z > 1$ (from graph)

(5) $Z > 1$ (from graph)

(6) same as (2) $Z < 1$

(7) $Z < 1$ (from graph)

(8) $Z_{CC} = \dfrac{3}{8}$

(9) $Z_{CC} = \dfrac{3}{8}$

(10) $Z = 1 + \left(b - \dfrac{a}{RT}\right)\times\dfrac{1}{V_m} + \text{.........neg. terms}$

$= Z > 1$